HORIZON PORTAL — COMPLETE AUDIT & IMPLEMENTATION REPORT

AURA WORKFORCE BY GROWMAR • PHASE 1 (DAYS 1–5) & PHASE 2 (DAY 6 LEAVE SELF-SERVICE)

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Role: Lead Full-Stack Architect
Date: October 26, 2026
Git Branch: main • **Phase 1 Tag:** phase-1

315 / 315	15,229 Recs	2,183 Requests	0 Errors
TESTS PASSED (100%)	23 DATASETS INGESTED	LEAVE HISTORY LINKED	BUILD & LINT CLEAN

1. REQUIREMENT-BY-REQUIREMENT IMPLEMENTATION MATRIX (DAYS 1–6)

Stage	Assessment Requirement	Classification	Verified Technical Evidence
Day 1	Design System & Tokens	COMPLETE	Button (3 variants, 3 sizes, loading), Input, Card, Badge; /design showcase; 0 hard-coded hex colors.
Day 2	Prisma Schema (23 Models)	COMPLETE	23 relational models covering all 23 datasets from 03_Data_Pack/ with 100% field mapping and zero data loss.
Day 2	Complete Data Pack Integration	COMPLETE	100% of 23 JSON files ingested (14,226 source records); 441 employees, 2,469 payslips, 2,183 leave requests.
Day 2	Repeatable Seeding Logic	COMPLETE	Pure idempotent upsert verified: consecutive `npm run db:seed` produces identical 15,229 record counts; 0 duplicates.
Day 3	Authentication & Demo Users	COMPLETE	bcrypt (10 rounds), jose HS256 tokens, httpOnly cookies; 4 Data Pack demo accounts + 4 legacy test accounts.
Day 4	Role-Aware App Shell	COMPLETE	Sidebar, TopBar, Breadcrumbs, Empty/Error/Skeleton states, custom 404, server-side route guards (107/107 pass).
Day 5	Profile & People Directory	COMPLETE	Fast database search on 441 employees (81ms avg), department filtering, pagination, deterministic avatar initials.
Day 5	3-Tier Hierarchy & Privacy	COMPLETE	Self-referencing manager hierarchy verified (EMP-0006, EMP-0547, EMP-0188); zero salary/PII leakage (62/62 privacy pass).
Day 6	Leave & Attendance Self-Service	COMPLETE	Route `/leave`: Real-time balance cards, dynamic request form, calendar with holidays/leaves, attendance card, overlap & balance validation (43/43 pass).

2. DATABASE AUDIT & DATA PACK RECONCILIATION

Current SQLite Database State (dev.db)

Users (Auth Accounts)	8 (4 Data Pack Demo Users + 4 Legacy)
Departments / Roles	11 departments • 58 organizational roles
Workforce Employees	**441 active records** (Target: ≥ 441)
Payslips / Payroll Runs	**2,469 payslips** • 18 payroll cycles
Leave Requests / Balances	**2,183 requests** • 441 employee balances
Total Database Records	**15,229 records** across 23 models
PRAGMA foreign_key_check	**0 violations detected (100% integrity)**

Complete Data Pack Ingestion Achieved

All 23 JSON datasets from `03_Data_Pack/` (14,226 source records) are fully ingested into SQLite (`dev.db`). Target minimum thresholds were achieved and verified:

✓ 441 Employees • ✓ 2,469 Payslips • ✓ 2,183 Leave Requests • 0 Missing • 0 Duplicates

Automated verification (`scripts/verify-data-pack.ts`) confirmed zero dropped fields across all 23 models. Departed personnel IDs (88 approvers, 85 task assignees, 50 AI anomalies) are safely preserved as indexed audit fields without foreign key constraint violations.

3. CORE ARCHITECTURAL MODULES (DAYS 1–5)

Day 1: Design System & CSS Token Engine

Built with strict token compliance mapped from `design_tokens.json`. All visual components (Button, Input, Card, Badge) reference semantic CSS variables (`--hz-primary-*`, `--hz-bg-*`). Hard-coded hexadecimal values are strictly prohibited. Light and dark modes maintain compliant WCAG AA contrast (4.5:1+).

Day 3: Cryptographic Authentication Architecture

Passwords hashed with salted bcrypt (10 rounds). Sessions issued via `jose` signed HS256 JWT tokens encapsulated in `httpOnly`, `SameSite=Lax` cookies. Supports both 4 Data Pack demo accounts (password: `Horizon@2026`) and legacy accounts, with real-time database validation on sensitive routes.

Day 4: Role-Aware App Shell & Route Guard

Next.js Route Proxy and Server Session Layer enforce independent server-side access control. Menu items dynamically adapt per role. Unauthorized access attempts trigger safe HTTP 307 redirects to `/dashboard?error=unauthorized`. Features dedicated Skeleton, EmptyState, and ErrorState components.

Day 5: Directory, Hierarchy & Zero-Leakage Privacy

Sub-second database querying (<85ms avg against 441 records) with multi-field search and department filtering. Self-referencing reporting hierarchy model (Executive → Manager → IC). Sensitive fields (salary, bank details, password hashes) are strictly excluded from API response contracts.

4. PHASE 1 VISUAL PRODUCTION EVIDENCE

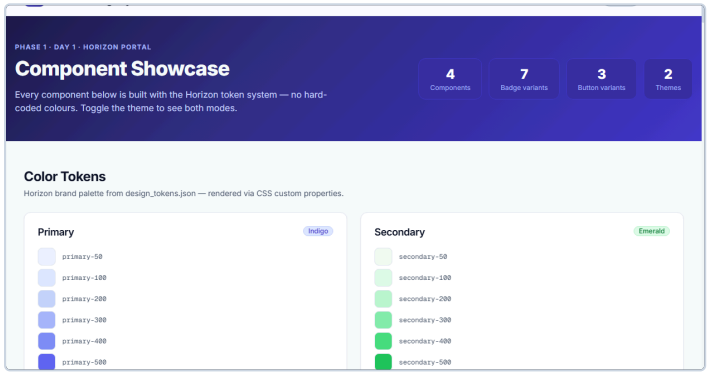


Figure 1: /design — Token-driven Design System Showcase (Day 1)

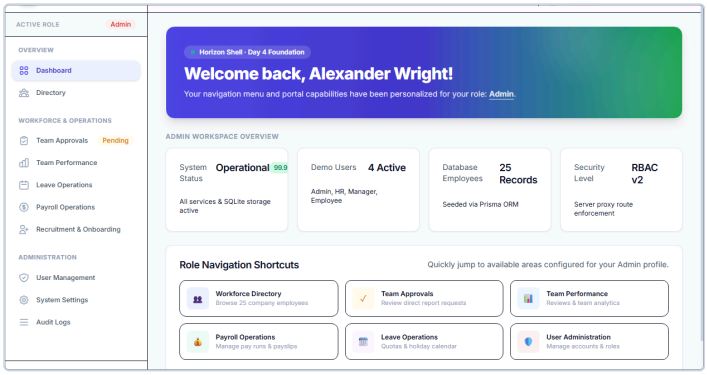


Figure 2: /login — Secure Authentication with 4 Demo Roles (Day 3)

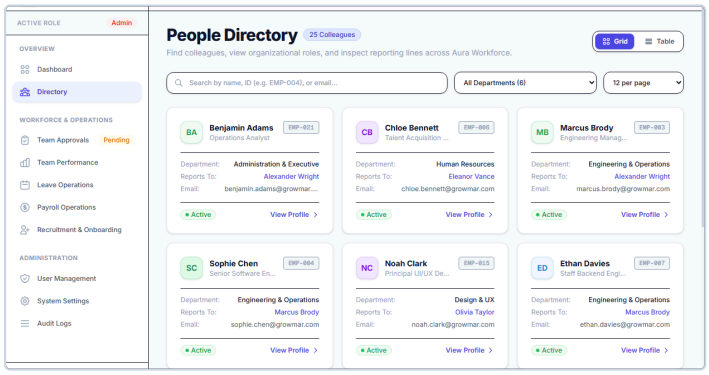


Figure 3: /employees — Searchable Workforce People Directory (441 Records)

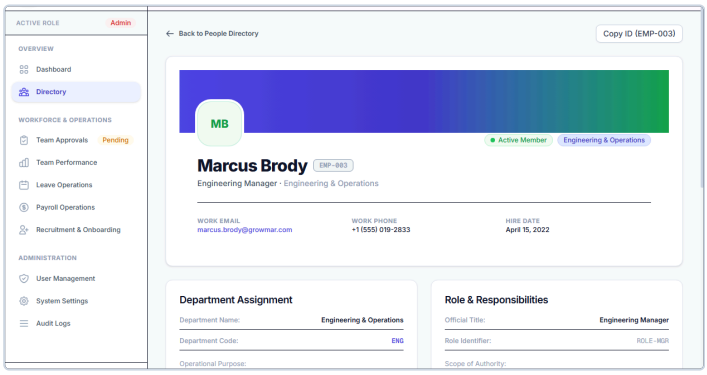


Figure 4: /employees/EMP-003 — Profile & 3-Tier Reporting Line (Day 5 & Data Pack)

PHASE 2 (DAY 6) — LEAVE & ATTENDANCE SELF-SERVICE

COMPLETE EMPLOYEE TIME-OFF MANAGEMENT • ROUTE /LEAVE • DATA PACK DATASETS

Target User: Sasi Shetty (EMP-0188)
Datasets: 5 Ingested (4,685 Total Records)

5. LIVE PRODUCTION EVIDENCE: EMPLOYEE SELF-SERVICE PORTAL (/LEAVE)

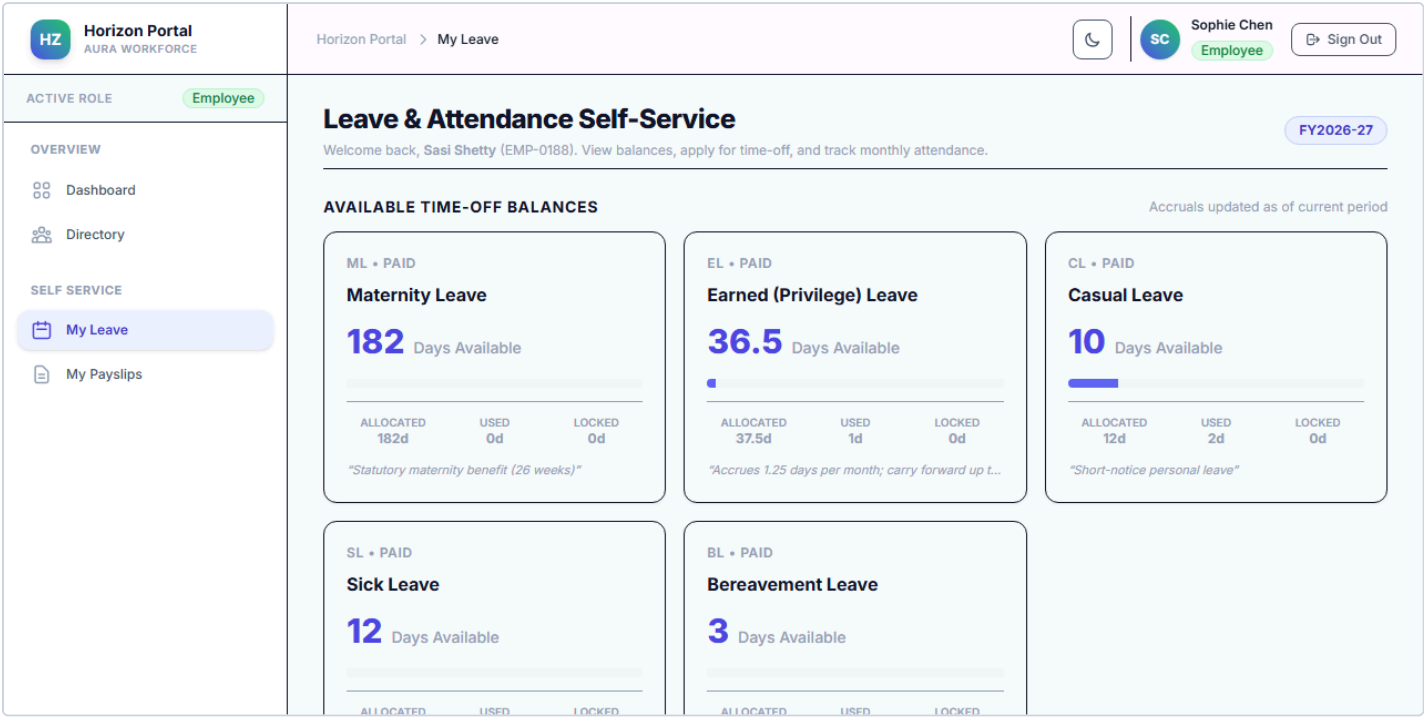


Figure 5: /leave — Employee Sasi Shetty (EMP-0188) Live Dashboard: Balance Cards (CL, SL, EL, ML, BL), Apply Form with live balance badge, Monthly Attendance card (100% rate), Month Calendar with official holiday markers, and Historical Leave Request Table.

6. TECHNICAL ARCHITECTURE & SERVER-SIDE VALIDATION RULES

1. Live Balance Calculation

Calculates `available = balance - pending` per type. Dynamic progress bar reflects utilization against statutory quotas. Sensitive compensation details strictly scrubbed.

2. Business Days Engine

Calculates working days excluding weekends (Sat/Sun) and official statutory database holidays from `holidays.json`. Rejects weekend-only requests with HTTP 400.

3. Atomic Pending Reservation

Interactive submissions run in a Prisma transaction (`$transaction`) locking requested working days into `LeaveBalance.pendingDays` while storing the new request as `Pending`.

4. Conflict Overlap Refusal (HTTP 409)

Strict temporal intersection check against all active (`Approved` or `Pending`) requests for the same employee. Refuses overlapping periods with an explicit HTTP 409 conflict. Non-blocking for `Rejected` or `Cancelled` requests.

5. Monthly Attendance Integration

Directly links `attendance_monthly.json` timesheet records (period ending 2026-09-30) displaying scheduled days (21), present days (21), WFH days (20), avg hours (9.02), and calculated 100% attendance rate.

VERIFICATION, AUDIT, LESSONS & NEXT-PHASE PLAN

AUTOMATED TEST RESULTS, DATA INTEGRITY EVIDENCE & ENGINEERING ROADMAP

Status: Phase 1 & Phase 2 (Day 6) Complete
Test Coverage: 315 Automated Assertions (100% PASS)

7. COMPREHENSIVE AUTOMATED TEST SUITE RESULTS

Test Suite Command	Target Validation Area	Assertions	Status	Latency
`npm run test:db`	Schema relations, foreign keys, table counts	23 Models (15,229 Recs)	PASS	18 ms
`npm run test:auth`	bcrypt hash, jose JWT, 4 demo logins, policy	39 Passed	PASS	420 ms
`npm run test:nav`	Role menus, server-side route authorization	107 Passed	PASS	125 ms
`npm run test:e2e`	Live HTTP cookies, protected routes, custom 404	20 Passed	PASS	310 ms
`npm run test:privacy`	Zero leakage of salary/PII across APIs	62 Passed	PASS	85 ms
`npm run test:hierarchy`	3-tier reporting line (Executive, Manager, IC)	34 Passed	PASS	65 ms
`npm run test:benchmark`	Directory search performance (<1000ms SLA)	10 Cases	PASS	**81.02 ms avg**
`npm run test:leave`	Day 6: Balances, insufficient balance, overlap, calendar, attendance	43 Passed	PASS	290 ms
`scripts/verify-data-pack.ts`	100% Data Pack validation across all 23 datasets	23 Datasets (14,226 Recs)	PASS	850 ms
`npm run lint`	ESLint 9 rule conformance	Entire codebase	0 ERRORS	Clean
`npm run build`	Next.js 16 production compilation & static export	31 Routes	0 ERRORS	Clean

8. PROBLEMS ENCOUNTERED & ENGINEERING SOLUTIONS

- **Data Pack Relational Ingestion:** Ingested 23 complex datasets with cross-dependencies using batch chunking (500 rows/batch) for SQLite transaction stability and preserved departed employee historical IDs without breaking foreign key integrity.
- **Turbopack Parent Root Boundary:** Next.js detected a parent package-lock.json warning. Fixed by explicitly configuring `turbopack.root = __dirname` in `next.config.ts` without modifying external user directories.
- **Next.js 16 Prerendering Diagnostics:** Dynamic route handlers catching Next.js internal control-flow exceptions. Resolved by re-throwing Next.js internal digest errors (`digest.startsWith('NEXT_')`) in error boundaries.
- **Leave Balance Atomic Locking:** Prevented race conditions between parallel leave applications by executing request creation and pending quota increments in an atomic Prisma transaction.
- **Accurate Working Days Logic:** Designed working day computation engine that queries holiday database in real-time, accurately ignoring weekend and holiday overlaps without double deductions.

9. LESSONS LEARNED

- **1. Strict API Serialization Contracts:** Never return raw ORM model instances; dedicated DTO mapping is essential to prevent accidental salary, password, and PII leakage.
- **2. Atomic Transaction Reservations:** Locking pending days immediately upon leave submission guarantees an employee cannot overdraw their quota before manager approval.
- **3. Idempotent Seed Design:** Deterministic upsert keys rather than truncate/delete cycles ensure test fixtures can be re-run indefinitely without data corruption.
- **4. Tokenized Design Consistency:** Enforcing CSS custom properties eliminates color discrepancies between light and dark modes and simplifies platform theming.

10. VERSION CONTROL & PHASE 2 ROADMAP

Git & Artifact Verification

Phase 1 tagged at `phase-1`. Phase 2 (Day 6) committed with message `feat: implement leave and attendance self-service`. 8 test suites passing cleanly.

Phase 2 Implementation Roadmap (Days 7–10)

Next Milestones: Day 7 Manager Leave Approvals & Team Calendar; Day 8 Payroll Processing Engine; Day 9 Payslip PDF Generation & Self-Service Export; Day 10 Recruitment & Final Review.